

THE RESULTS OF YOUR ANALYSIS

PREPARED FOR

Anthony Brown

DATE OF INJURY : 2/1/2020

DATE OF ANALYSIS : 2/18/2020

DATE OF IMAGES : 2/11/2020

REFERRING DOCTOR :

Buffalo Chiropractic & Acupuncture, PLLC

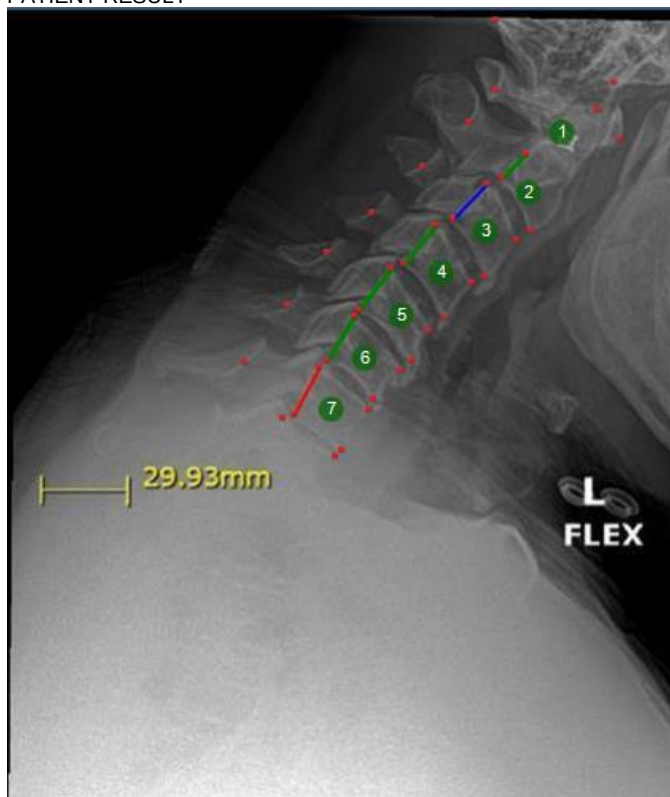
This report contains important information concerning structural changes that can be affecting your overall level of health and well-being. Spinal stability is a basic requirement for the protection of your nervous structures and the prevention of early mechanical deterioration of your spinal component

Instability is generally considered to be a global increase in the movements associated with the occurrence of back, neck, and/or nerve root pain. Damage to any spinal structure produces some degree of spinal instability.

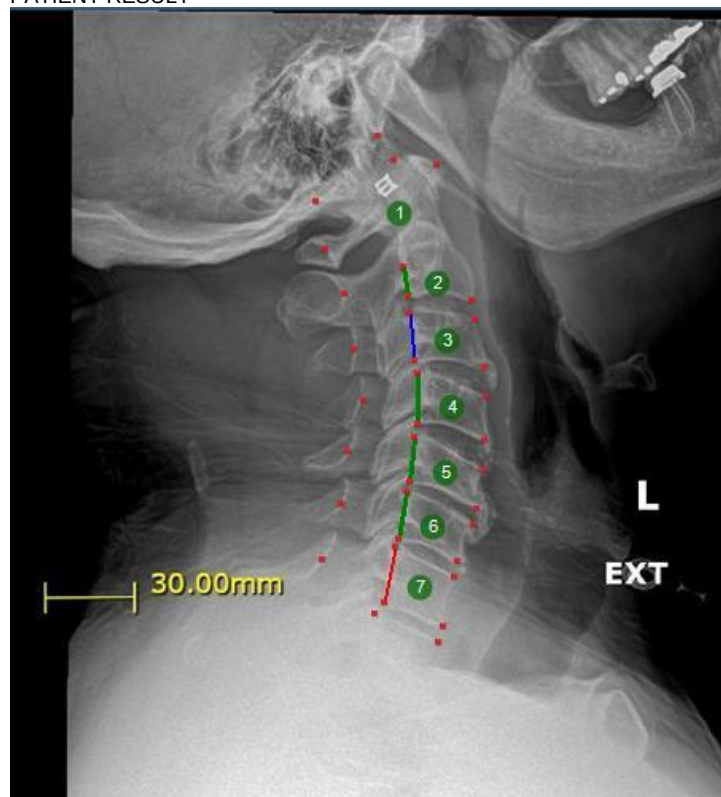
This computer aided digital analysis is an overview of your current level of spinal stability.

Cervical Flexion / Extension

PATIENT RESULT



PATIENT RESULT



This RED line denotes the location of the posterior longitudinal ligament (PLL) and surpasses normal acceptable segmental motion indicating the location of posterior longitudinal ligament laxity.

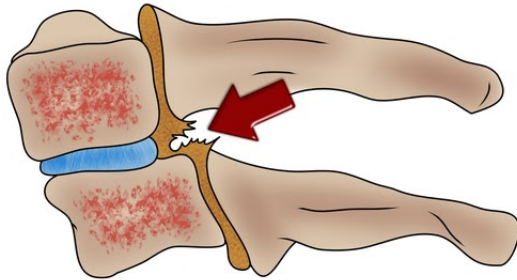
This GREEN line denotes the location of the posterior longitudinal ligament (PLL) with normal alignment to the surrounding vertebrae indicating no significant ligamentous laxity.

A blue line denotes a segmental motion that is clinically significant.

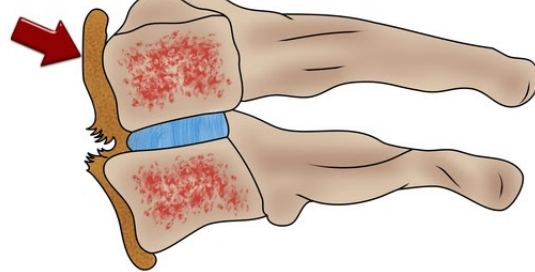
Alteration of motion segment integrity is defined as abnormal translation or angular motion between two adjacent vertebrae. This motion is measured using lateral flexion and extension x-rays. Translational motion segment integrity is assessed on flexion and extension imaging by comparing and measuring the position of the posterior edge of adjacent vertebral bodies in each position. Alteration of angular motion segment integrity is assessed by comparing the angles formed by the superior endplates of adjacent vertebral bodies (adjacent motion segments). These spinal translation and angulation are measured using a process referred to as Digital Radiographic Mensuration Analysis (DRMA). SpineTech Pro software captures articular motion to quantify spinal motion pathology and identify the presence and location of ligamentous laxity (ICD-9 728.4 and ICD-10 M24.28). The World Health Organization has rated this method of analysis as "ESTABLISHED".

Cervical Flexion / Extension - Translational Motion Results

Posterior Longitudinal Ligament Injury



Anterior Longitudinal Ligament Injury



Vetebral Level	% Anterior	% Posterior	% Range Clinically Significant	% Ratable	Impression
C2-C3	4.2	5.7	5.8-19.9	> 20	Normal
C3-C4	11.2	3.4	5.8-19.9	> 20	Clinically Significant
C4-C5	2.7	4.6	5.8-19.9	> 20	Normal
C5-C6	1.8	3.8	5.8-19.9	> 20	Normal
C6-C7	4.1	1.2	5.8-19.9	> 20	Normal

Cervical Spine Translational AOMSI

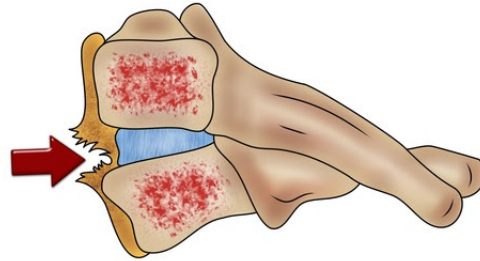
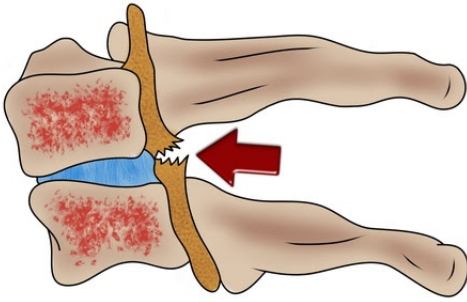
"Cervical spine AOMSI is defined using flexion/extension x-rays. A diagnosis of AOMSI in the cervical spine by translation measurements requires greater than 20% anterior or greater than 20% posterior relative translation of one vertebrae on another, on flexion or extension radiographs, respectively" *AMA Guides to the Evaluation of Permanent Impairment 6th Edition*

Additional understanding of Alteration of Motion Segment Integrity (AOMSI) can be found in an article titled "Biomechanical Analysis of Clinical Instability in the Cervical Spine" White, et al., Clin Ortho Relat Res, 1975;(109):85-96. White et al supports that when the motion segment stayed intact i.e., less than ratable translation and/or angular motion it could still be clinically significant even though it does not reach the AMA Guides threshold to being ratable. These "Clinically Significant" ratings are injuries just under the threshold of severe to very severe. These patients could be at increased risk of future catastrophic injury resulting from a minimal force event due to existing non-ratable ligamentous injury.

Cervical Flexion / Extension - Angular Motion Results

FLEXION Posterior Longitudinal Ligament Injury

EXTENSION Anterior Longitudinal Ligament Injury



Bio-mechanical measurements are calculated using the right hand Cartesian coordinate system where a preceding "-" sign represents a posterior translation of linear motion and a "-" preceding angular motion indicates rotational motion into extension

Vertebral Level	Flexion	Clinically Significant	Rateable Limit	Comments
C2-C3	1.33	7-11°	> 11°	Normal
C3-C4	3.86	7-11°	> 11°	Normal
C4-C5	2.24	7-11°	> 11°	Normal
C5-C6	6.3	7-11°	> 11°	Normal
C6-C7	3.19	7-11°	> 11°	Normal

Cervical Spine Rotational AOMSI

"Cervical spine AOMSI is defined using the lateral flexion x-rays. A diagnosis of AOMSI based on angular motion of more than 11° greater than each adjacent level on the flexion radiograph. *AMA Guides to the Evaluation of Permanent Impairment 6th Edition*

Additional understanding of Alteration of Motion Segment Integrity (AOMSI) can be found in an article titled "Biomechanical Analysis of Clinical Instability in the Cervical Spine" White, et al., Clin Ortho Relat Res, 1975;(109):85-96. White et al supports that when the motion segment stayed intact i.e., less than ratable translation and/or angular motion it could still be clinically significant even though it does not reach the AMA Guides threshold to being ratable. These "Clinically Significant" ratings are injuries just under the threshold of severe to very severe. These patients could be at increased risk of future catastrophic injury resulting from a minimal force event due to existing non-ratable ligamentous injury.

Alteration Of Motion Segment Integrity (AOMSI) Analysis

patient's NAME: Anthony Brown

DATE OF BIRTH: 02/19/1963

DATE OF INJURY: 02/01/2020

DATE OF ANALYSIS: 02/18/2020

Digital Radiographic Mensuration Analysis (DRMA)

DRMA analysis is a digital radiologic analysis necessary to determine AOMSI and alignment and ligamentous stability after trauma. DRMA technology provides an accurate diagnosis. In addition, DRMA follows the strict criteria of the **AMA Guides to the Evaluation of Permanent Impairment**, 5th Edition in calculating impairment from lateral Cervical Flexion / Extension flexion and extension radiographs. (Cocchiarella Land Andersson GBJ. AMA Press 2001). This is not a radiology report, it is a distinct and separate biomechanical analysis and the findings should be clinically correlated by the treating physician.

Lateral Flexion/Extension Views of the Cervical Flexion / Extension Spine Were Obtained and Reviewed for AOMSI:

- Cervical Flexion / Extension Angular Motion is below ratable threshold.
- The translation motion segment integrity is not ratable, but is clinically significant at C3-C4 at 11.2 Percent.

IMPRESSIONS :

- Translation variation of Motion Segment Integrity is clinically significant, but not ratable, at C3-C4 at 11.2 percent and this may add to this patient's total impairment.

Buffalo Chiropractic & Acupuncture, PLLC

Analysis Notes:

Measuring Translation :

In any of the spinal segments, lateral flexion and extension x-rays are analyzed. In the presence of true loss of motion segment integrity, the upper vertebrae will move forward on the lower vertebrae in flexion and move backwards on the lower vertebra in extension. Thus, the line along the back cortex of the upper vertebra will be anterior to the line along the posterior cortex of the lower vertebra on the flexion x-ray and posterior to it on the extension x-ray. The measurable angle is the sum of the two measurements.

If the total of the two measurements exceeds 4.5 mm in the lumbar spine or 3.5 mm in the cervical spine, the definition of loss of motion segment integrity is met. In the event that the line along the posterior cortex of the upper vertebra remains anterior to the line along the posterior cortex of the lower vertebra on the extension view, then consider only the distance between the lines on the flexion view, as it would not be appropriate to subtract the distance on the extension view from that on the flexion view.

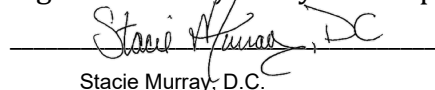
Measuring Angular Motion:

Lateral flexion and extension x-rays are used. Lines are drawn along the superior cortex of the two vertebrae adjacent to the level in question on both the flexion and the extension x-rays and the angles where these lines intersect measured on both views. The angle on the extension view is then subtracted from the angle on the flexion view. Normally, the angle on the flexion view will be positive and the angle on the extension view will be negative. The net result will be that the negative extension angle will be subtracted from the positive flexion angle resulting in a net addition of the two angles. ($+8 - (-18) = +26$). If the angle is greater than 15 degrees at L1-2, L2-3 or L3-4 or greater than 20 degrees at L4-5 or greater than 25 degrees at L5-S1, the definition of loss of motion segment integrity is met.

In the cervical spine, the method is different. A lateral flexion view only is taken. Lines are drawn along the inferior cortex of the two vertebrae above and below the level in question. Three angles are measured, the angles at the levels above and below the level in question and the angle at the level in question. If the flexion angle at the level in question is more than 11 degrees greater than either of the angles at the levels above or below, the definition of loss of motion segment integrity is met. The figure in the 5th Edition is correct, but the text under the figure suggests that both flexion and extension films are necessary when actually only a flexion view is necessary.

These are the only definitions of loss of motion segment integrity.

Ligament Laxity Analysis Completed by:


Stacie Murray, D.C.